Information Security

Unit- 2.3
Symmetric Encryption

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TU

Principle of Block Cipher

- A block cipher is a type of symmetric-key encryption algorithm that operates on fixed length group of bits or blocks, and transform them in to ciphertext using a cryptographic key.
- The principle of block cipher can be summarized as follows -

1. **Key Generation :**
   A secret key is generated by the sender and shared with the receiver through a secure channel.
2. **Substitution :**
   Each block of plaintext is substituted with a corresponding block of ciphertext using a mathematical function that is depend on key. This process is known as substitution or confusion.
3. **Permutation :**
   The order of block is rearranged using a mathematical function that is also dependent on the key. This process is also known as permutation or diffusion.

Principle of Block Cipher

4. **Iteration/ Rounds :**

   The substitution and permutation steps are repeated for a fixed number of round, to increase the security of the cipher.

5. **Decryption :**

   To decrypt the ciphertext, the process is reversed by performing the inverse permutation and substitution using the same key.

Confusion and Diffusion

- The term confusion and diffusion were introduced by Claude Shannon in 1949 AD.
- Shannon's concern was on to prevent cryptanalysis, based on statistical analysis. Assume attacker has some knowledge of the statistical characteristics of the plaintext. For example in a message, the frequency distribution of the various letters may be known. If these statistics are in any way reflected in the ciphertext, the attacker may be able to deduce the encryption key.
- Thus Shannon suggested two methods for frustrating the attacker:

    1. **Confusion**
    2. **Diffusion**

1. Confusion

- Confusion refers to making the relationship between the key and ciphertext as complex as possible.
- In confusion, each bit of the ciphertext should depend on the entire key, changing one bit of the key should change the ciphertext completely.

2. Diffusion

- The idea of diffusion is to hide the relationship between the ciphertext and the plaintext.
- Diffusion means that if we change a single bit of the plaintext then more than half of the bits in the ciphertext should change.
- Similarly, if we change one bit of ciphertext then more than half of the plaintext bits should change.

CONFUSION vs DIFFUSION

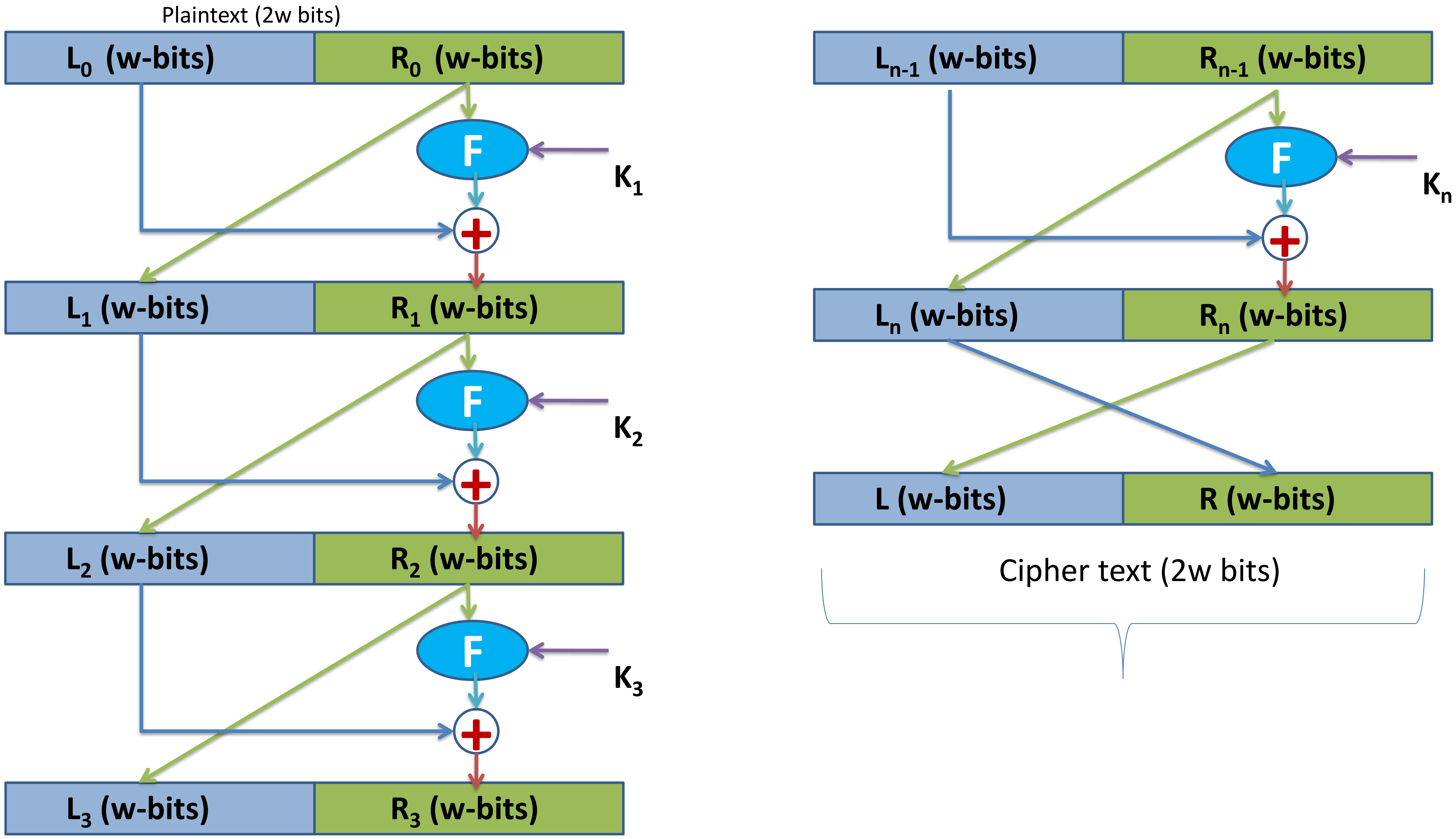
	Confusion	Diffusion
Meaning	Hides the relationship between the key and ciphertext.	Hides the relationship between the plaintext and ciphertext.
Main Purpose	Makes the key difficult to guess.	Hides patterns in the original message.
Method Used	Usually done by substitution.	Usually done by transposition / permutation.
If One Bit Changes	Changing one bit of the key changes many ciphertext bits.	Changing one bit of plaintext changes many ciphertext bits.
Focus	Key security.	Data security.
Result	Attacker cannot understand how the key works.	Attacker cannot detect message patterns.

Feistel Cipher Structure

- The Feistel cipher, introduced by Horst Feistel , is a design model or structure used to built various symmetric block ciphers such as DES.
- It uses same key for encryption and decryption.

- Feistel cipher structure encrypts plaintext in several rounds, where it applies substitution and permutation to the data.
- Each round uses a different key for encryption and that same key is used for the decryption process.

Feistel Cipher Structure

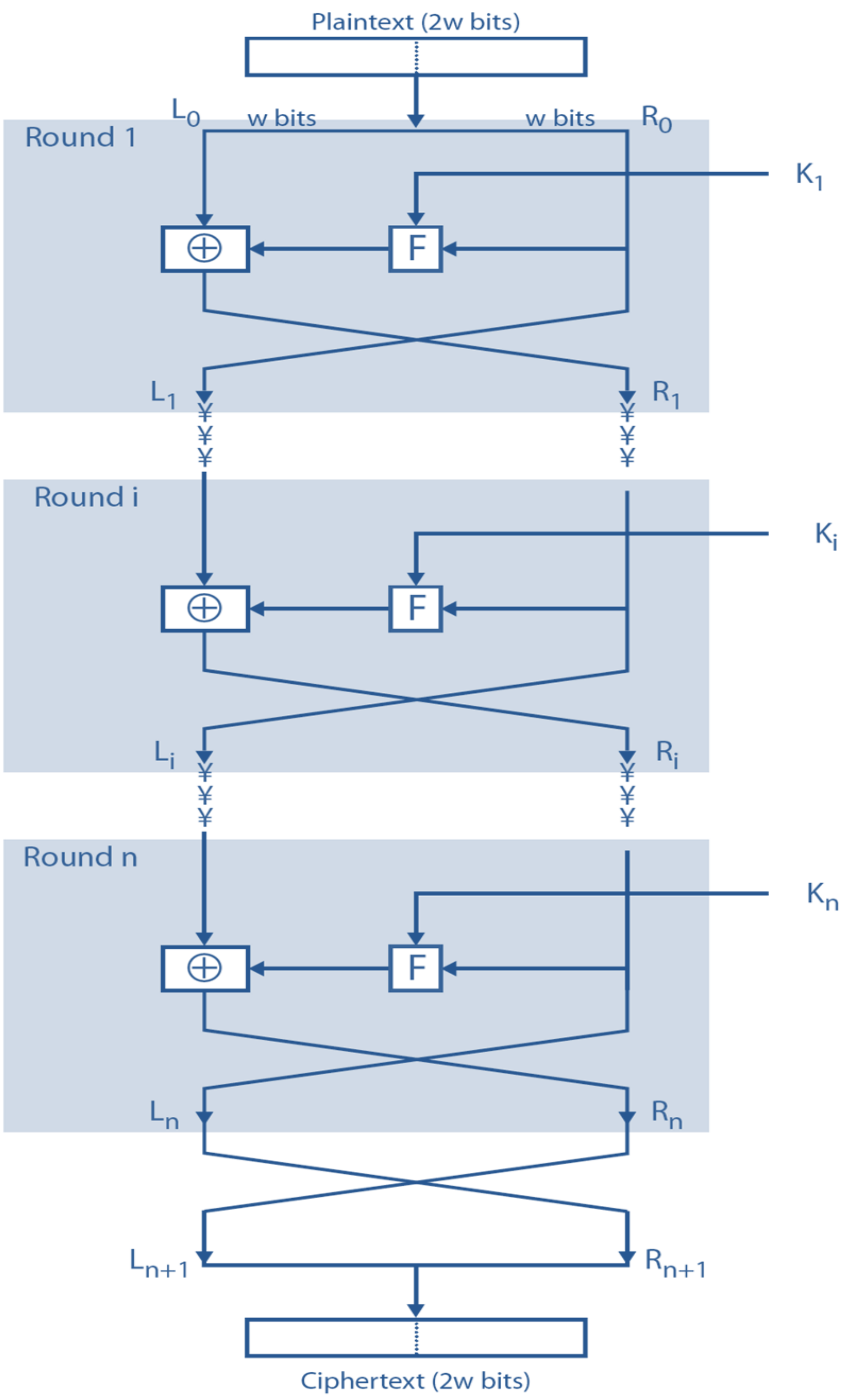


Encryption :

- The encryption process proposed by Feistel cipher is depicted in figure above.

- The inputs to the encryption algorithm are a plaintext block of length 2w bits and a key K.

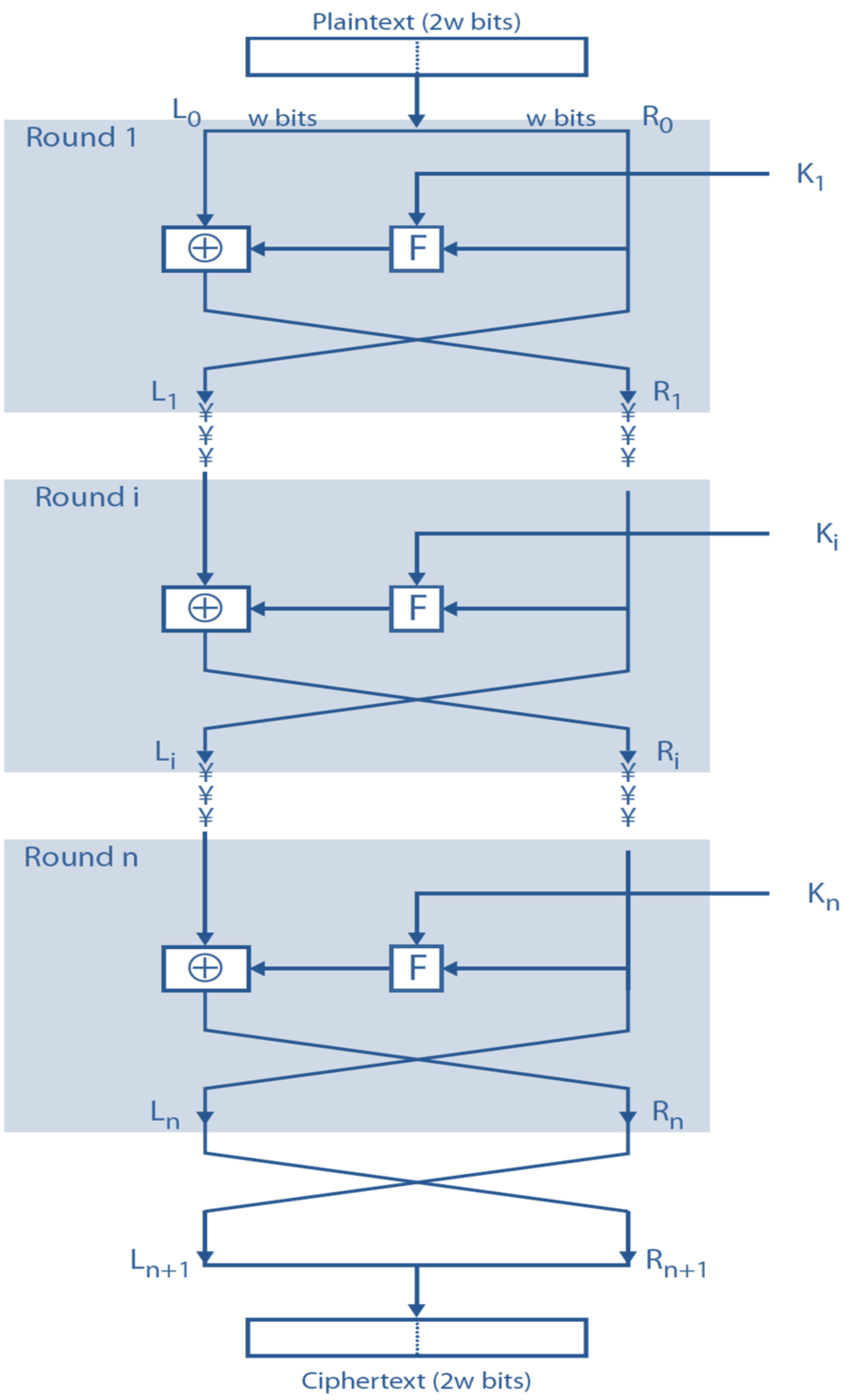
- The plaintext block is divided into two halves, L_0 and R_0 . The two halves of the data pass through n rounds of processing and then combine to produce the ciphertext block. Each round i has as inputs L_{i-1} and R_{i-1} derived from the previous round, as well as a subkey K_i derived from the overall K. In general, the subkeys K_i are different from K and from each other.



- To generate left half of the next round (L_{i+1}), the current right half R_i is assigned to it.
- To generate right half of the next round (R_{i+1}), the current right half R_i undergoes the following steps -

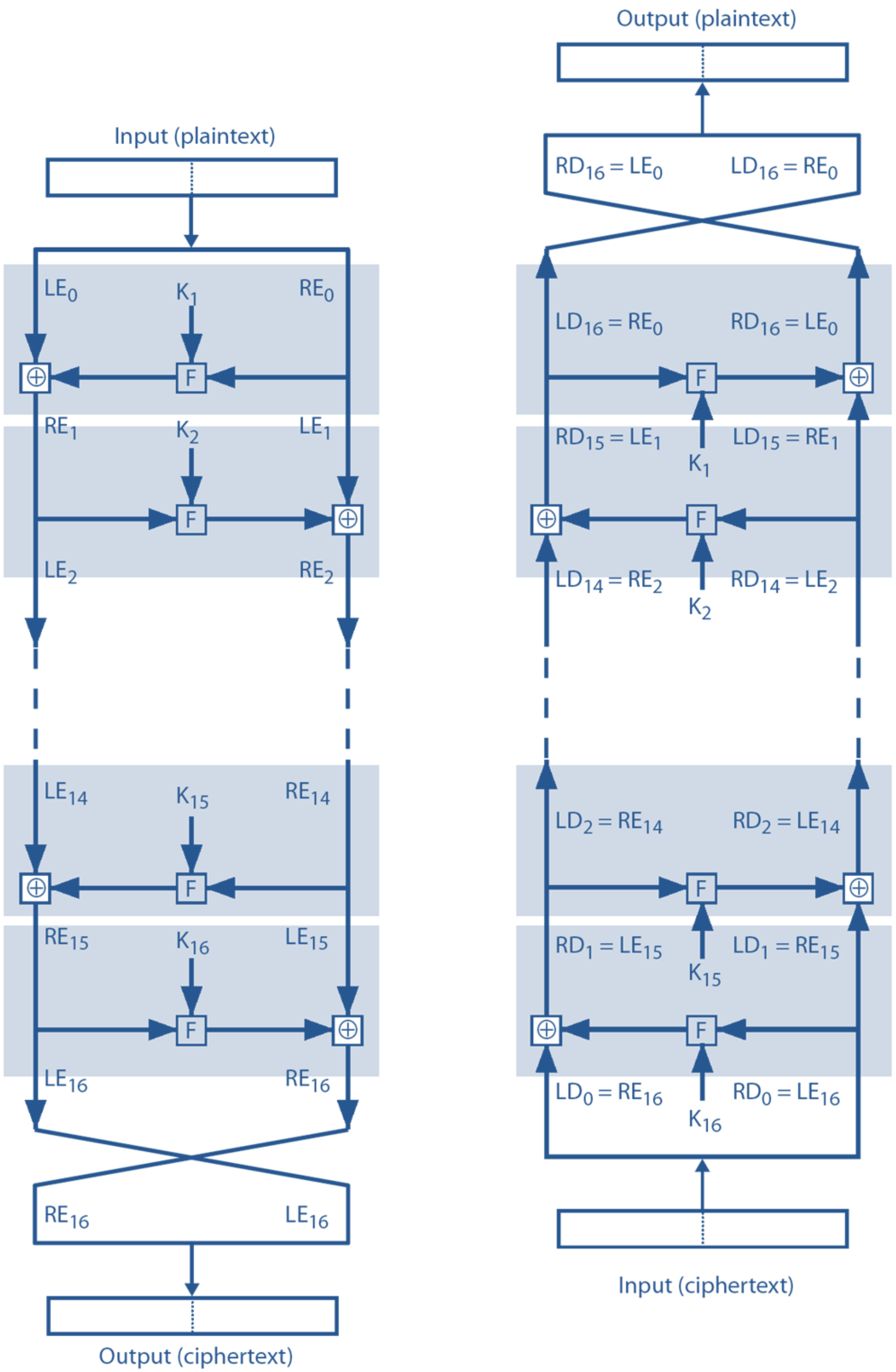
    1. R_i and the subkey K_i is passed through an encryption function.
    2. The result of (1) is X-ORed with the left half of the current round L_i .
    3. The result from (2) is assigned to the right half of the next round i.e R_{i+1} .

- The left and right half of data obtained after n rounds of execution is swapped again before concluding the Feistel cipher.



Decryption:

- The decryption process uses a similar procedure; ciphertext is fed to the algorithm and the exact steps are followed. The only difference is that the key used in decryption process follow a reverse order of that used in encryption process.



Design Features of Feistel Cipher

- Feistel cipher structure has the following five components-

1. **Block Size :**
   The larger the size of the block, the more secure and comlex the block cipher is. However, a large block size reduces the execution speed of the encryption and decryption process.

2. **Key Size :**
   Large size key increases the security of the block cipher. However, it also makes the encryption and decryption process slow.

3. **Number of Rounds :**
   The greater the number of round, n, used for the encryption/decryption process, the higher the complexity.

Design Features of Feistel Cipher

4. **Sub-key Generation Algorithm :**
   Complex algorithm make it difficult for intruders to crack the key.

5. **Encryption Function :**
   Complex function enhance the security of the block cipher, making them difficult to crack.

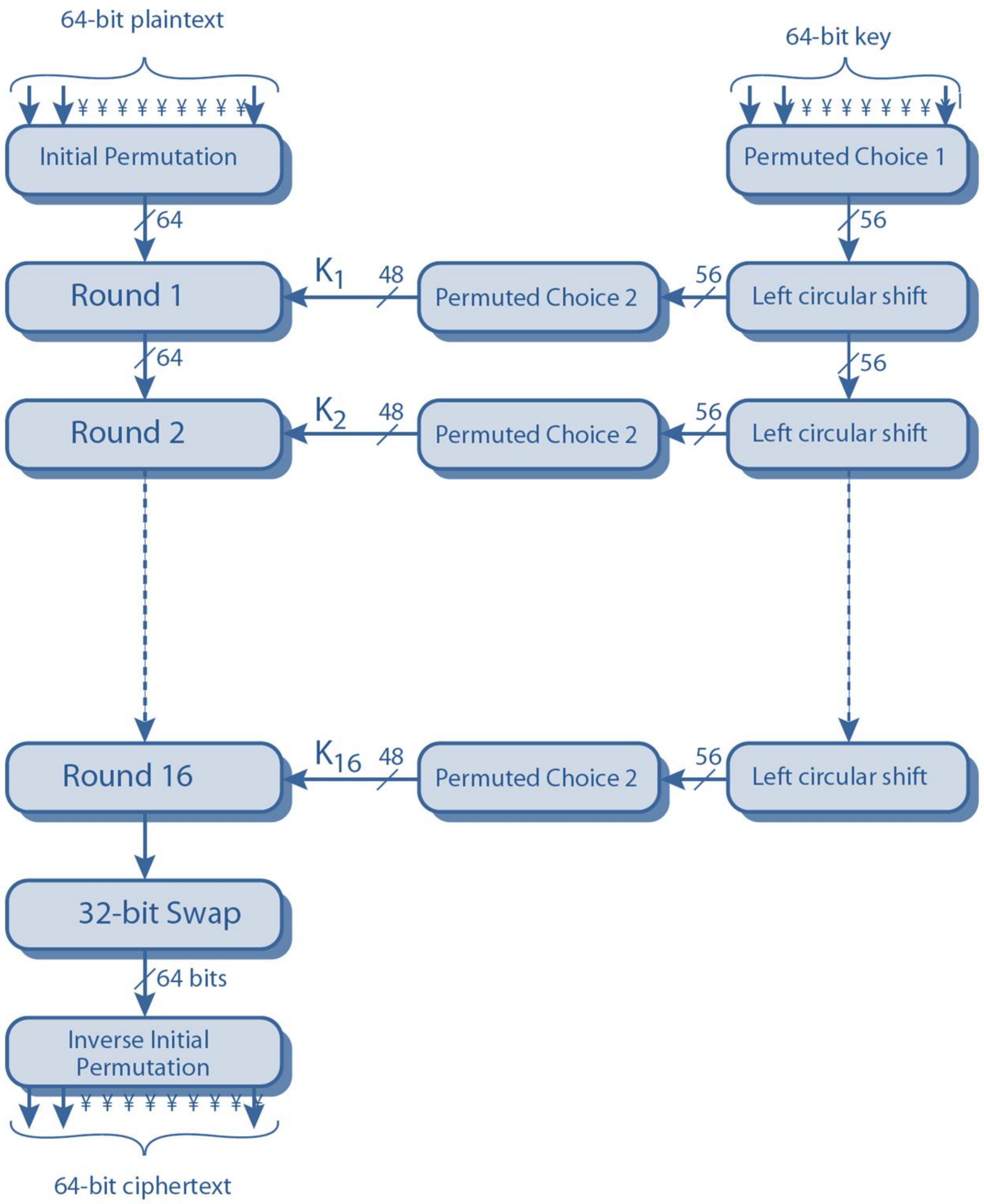
Data Encryption Standard (DES)

- DES is a symmetric key block cipher algorithm that was developed by IBM in 1970 AD (Lucifer cipher) and later adopted by the US government as a standard for protecting sensitive data.

Features or key points of DES:

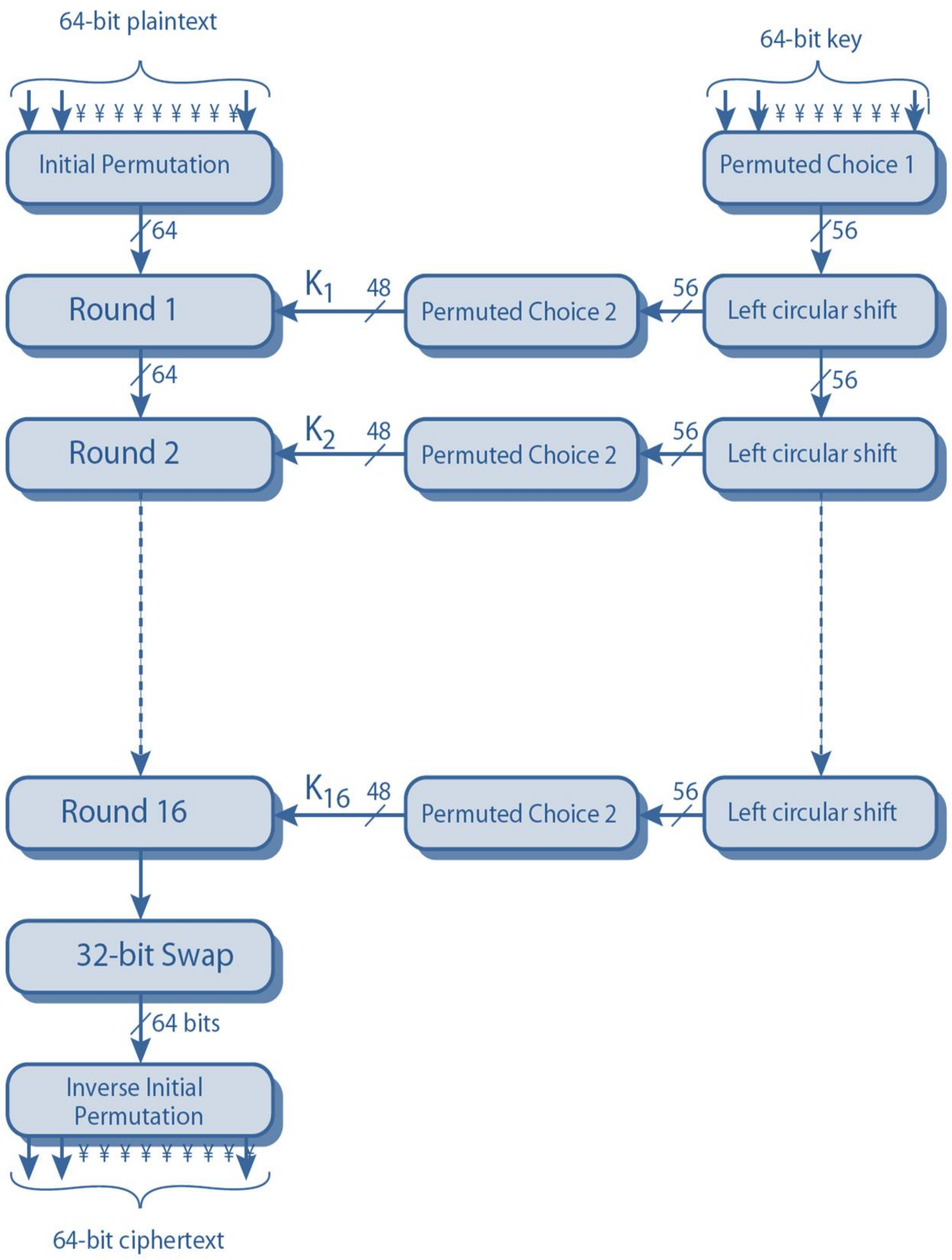
1. Key Length = **56 bit**
2. Block Size = **64 bit**
3. Cipher Type = **Symmetric Block Cipher**
4. Developed = **1977 AD**.
5. Possible key = $\mathbf{2^{56}}$
6. Round = **16**
7. Sub-key = **16 sub-keys with 48-bit**
8. S-Box = **8 (6 to 4)**

Data Encryption Standard



Encryption:

- The overall scheme for DES encryption is illustrated in Figure.
- As with any encryption scheme, there are two inputs to the encryption function: the plaintext to be encrypted and the key. In this case, the plaintext must be 64 bits in length and the key is 56 bits in length.
- Looking at the left-hand side of the figure, we can see that the processing of the plaintext proceeds in three phases.
- First, the 64-bit plaintext passes through an initial permutation (IP) that rearranges the bits to produce the permuted input.

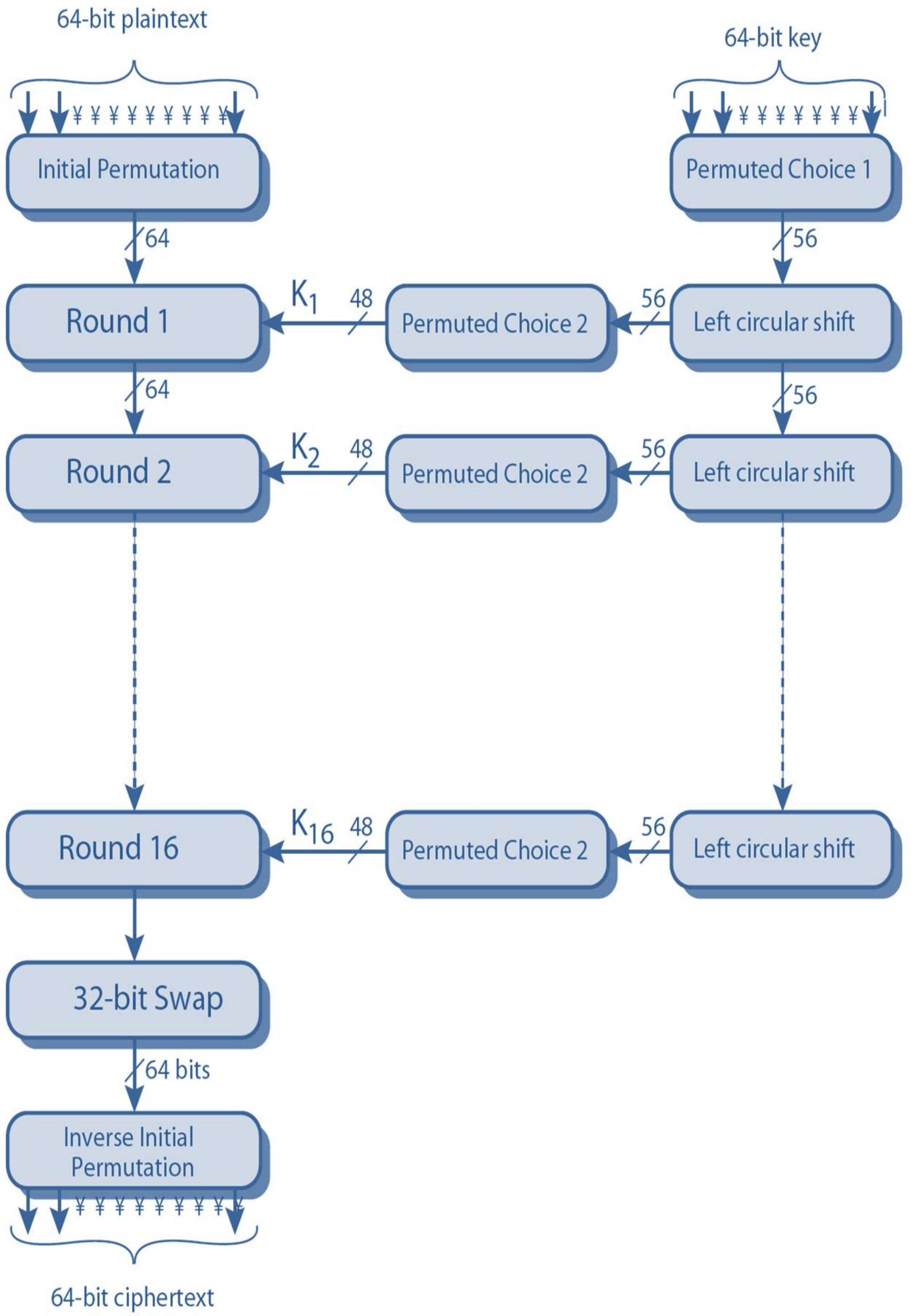


➢ This is followed by a phase consisting of sixteen rounds of the same function, which involves both permutation and substitution functions. The output of the last (sixteenth) round consists of 64 bits that are a function of the input plaintext and the key. The left and right halves of the output are swapped to produce the pre-output.

➢ Finally, the pre-output is passed through a permutation [IP^{-1}] that is the inverse of the initial permutation function, to produce the 64-bit ciphertext.

➢ The right hand portion of figure shows the way in which the 56-bit key is used. Actually the key is initially 64-bit and a parity dropper converts 64-bit to the 56-bit key. Initially, the key is passed through a permutation function, then for each of the 16 round a sub-key K_i is produced by a combination of left circular shift and a permutation.

➢ The permutation function is the same for each round, but a different sub-key is produced because of the repeated shifts of the key bits.

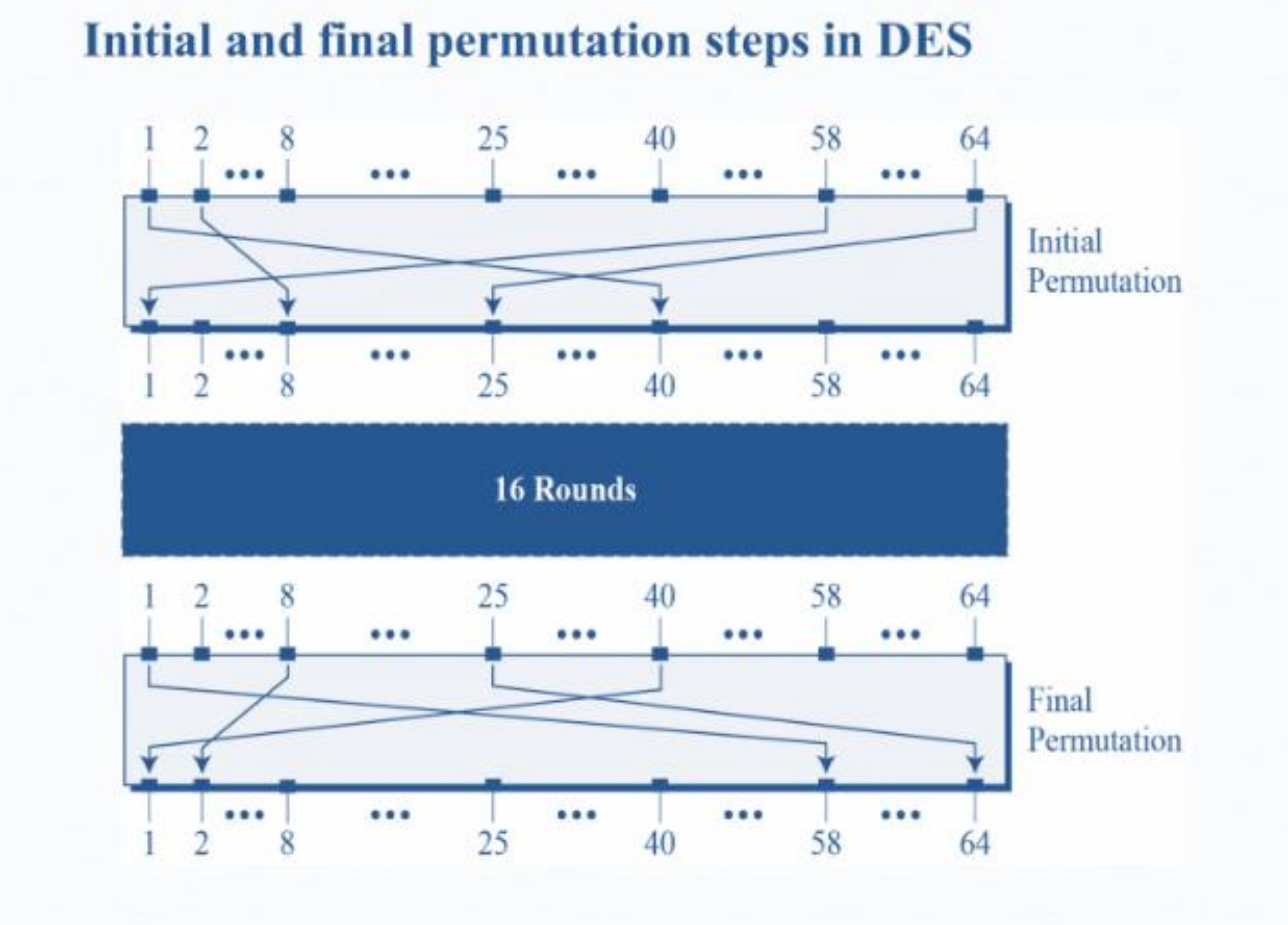


Decryption:

- As with any Feistel cipher decryption uses the same algorithm as encryption, except that the application of a subkeys is reversed. Additionally the initial and final permutation are reversed.

Initial and Final Permutation:

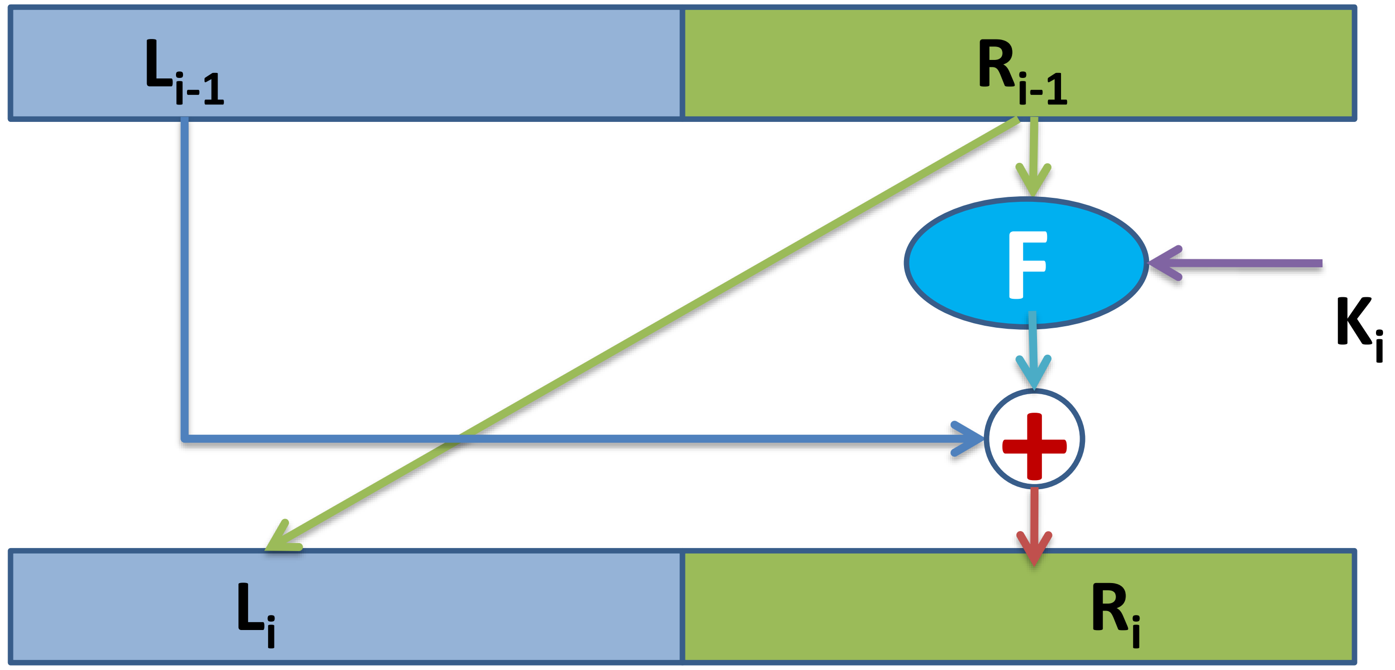
➢ The initial and final permutations are straight Permutation boxes (P-boxes) that are inverses of each other. The initial and final permutations are shown as follows –



Initial and final permutation tables

Initial Permutation	*Final Permutation*
58 50 42 34 26 18 10 02	40 08 48 16 56 24 64 32
60 52 44 36 28 20 12 04	39 07 47 15 55 23 63 31
62 54 46 38 30 22 14 06	38 06 46 14 54 22 62 30
64 56 48 40 32 24 16 08	37 05 45 13 53 21 61 29
57 49 41 33 25 17 09 01	36 04 44 12 52 20 60 28
59 51 43 35 27 19 11 03	35 03 43 11 51 19 59 27
61 53 45 37 29 21 13 05	34 02 42 10 50 18 58 26
63 55 47 39 31 23 15 07	33 01 41 09 49 17 57 25

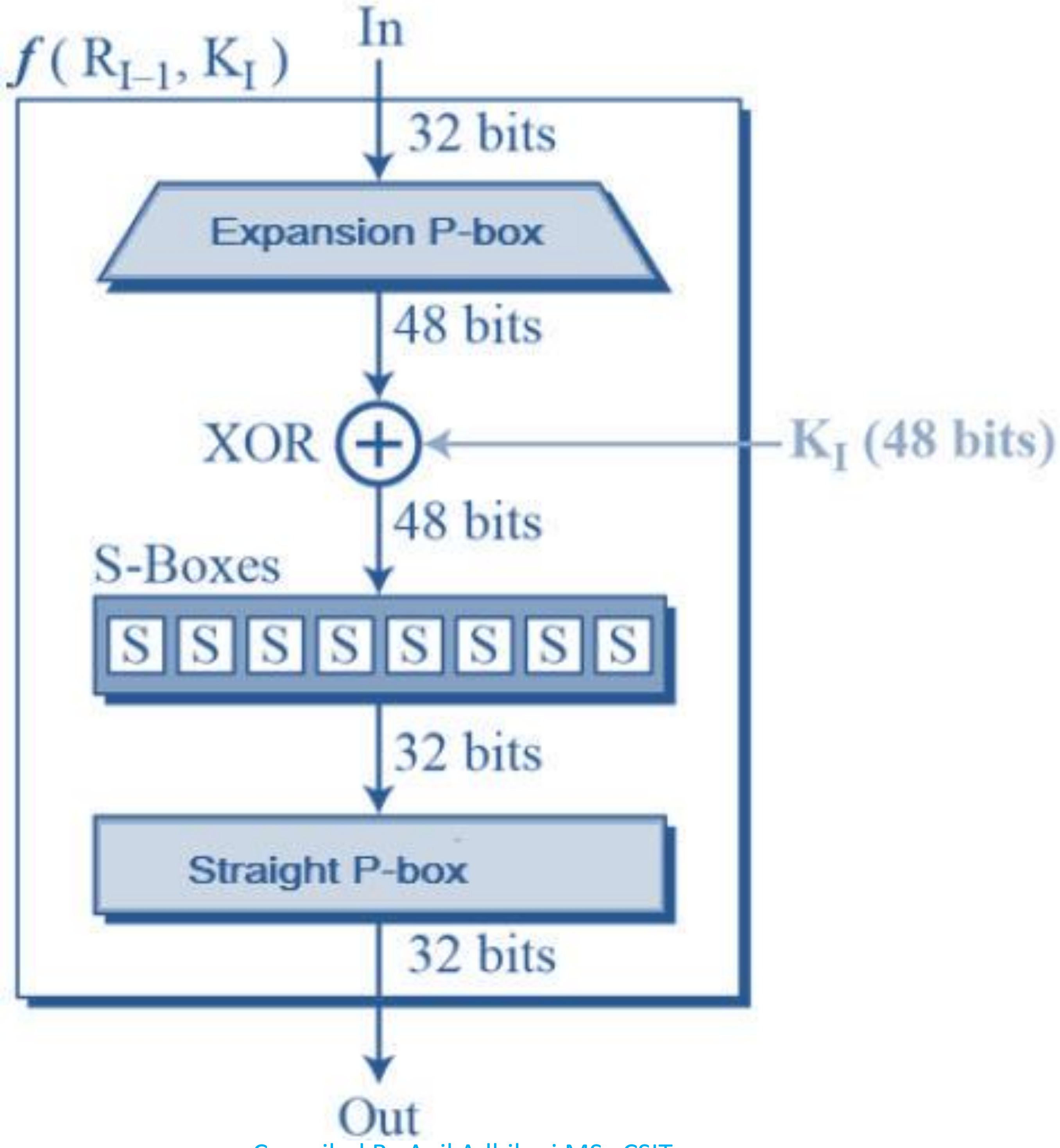
One Round of DES



- One round of DES encryption is shown in figure above.
- Each L_{i-1} and R_{i-1} is a 32-bit in length.
- The function F: $\{0,1\}^{32} \times \{0,1\}^{48} \rightarrow \{0,1\}^{32}$, takes 32-bit input string (right half of current state) and a round key or sub key and gives 32-bit output.
- This 32-bit output value is again X-Ored with the left half of previous state (L_{i-1}) and the resulted 32-bit is assigned to Right half of the current state (R_i).
- Similarly, Right half of the previous state is as it is assigned to left half of current state.

F-function of DES

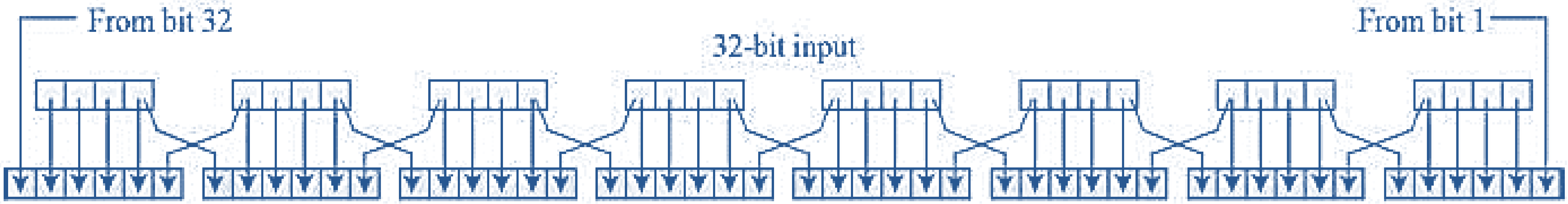
- The heart of this cipher is the DES function, f. The DES function applies a 48-bit key to the rightmost 32 bits to produce a 32-bit output.



F-function of DES

1. Expansion P-Box :

- Since right input is 32-bit and round key is a 48-bit, so first we need to expand right half input to 48-bit. The expansion logic is graphically depicted as below-



- In tabular form it can be illustrated as -

32	01	02	03	04	05
04	05	06	07	08	09
08	09	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	31	31	32	01

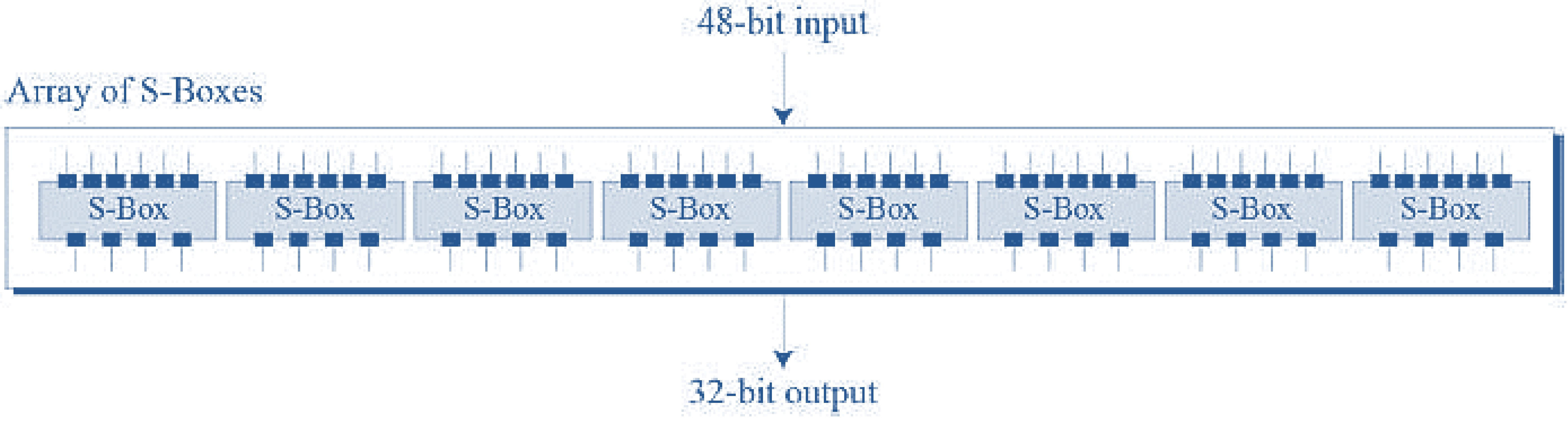
F-function of DES

2. X-OR (Whitener) :

- After the expansion permutation, DES does X-OR operation on the expanded right section and the round key.

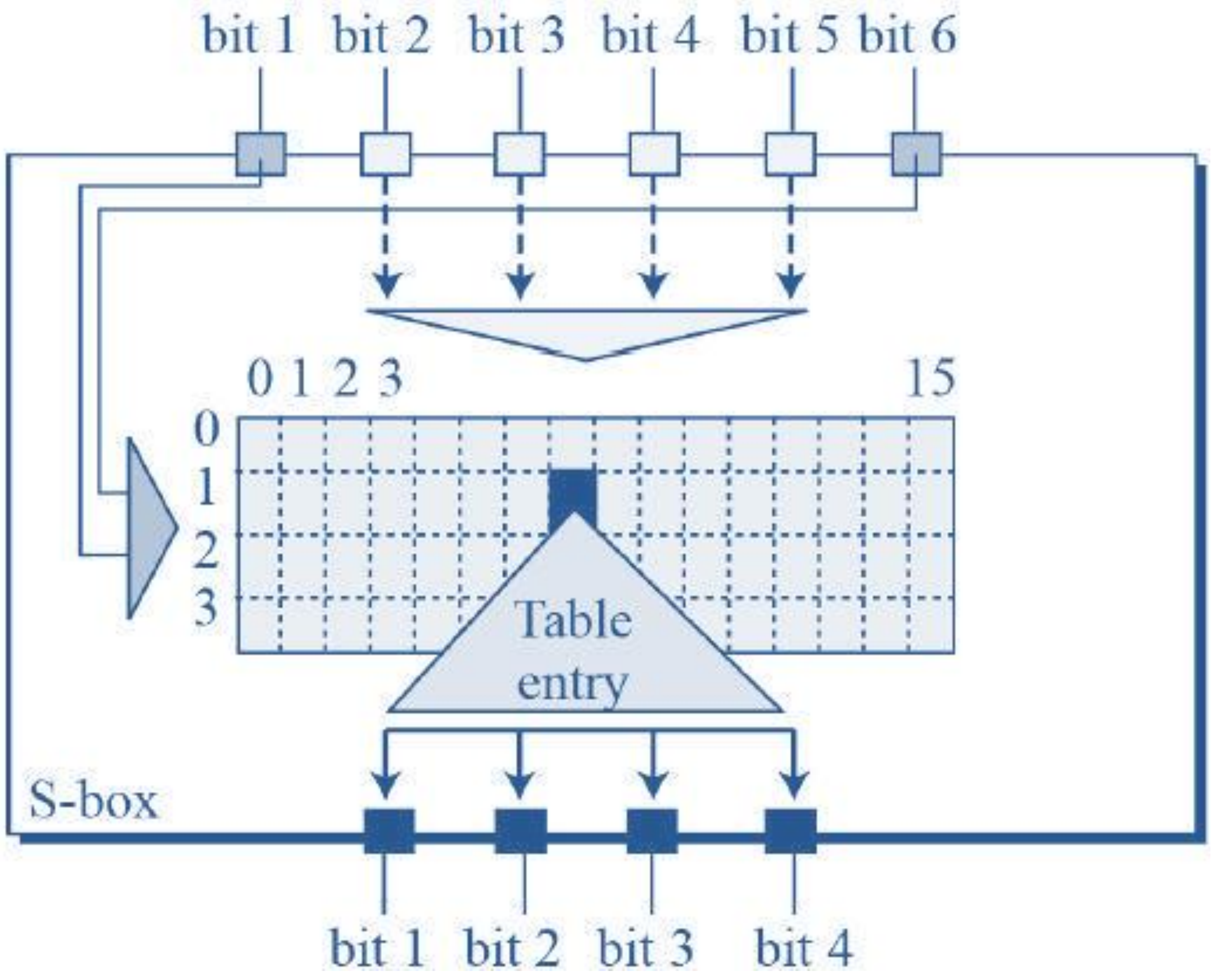
3. Substitution Boxes :

- The S-boxes carry out the real mixing (confusion).
- DES uses 8 S-boxes, each with a 6-bit input and a 4-bit output. Refer the following illustration –



F-function of DES

- The S-Box rule is illustrated below-



- Combination of first and sixth bit gives row information.
- Combination of 2nd , 3rd , 4th and 5th bit gives column information.
- There are total 8 S-Box tables.

F-function of DES

4. Straight P-Box :

- The 32-bit output of S-boxes is then subjected to the straight permutation with predefined rule or table.

Key Generation for DES

- The round-key generator creates sixteen 48-bit keys out of a 56-bit cipher key. The process of key generation is depicted in the figure.

- Initially we have a 64 bit key which is transformed into 56-bit key by using parity drop. The parity dropper drop every 8th bit of initial key. The round key generator then creates 16, 48-bit keys out of the 56-bit key.

- Here the 56-bit key is divided into two halves, each of 28 bits. These halves are circularly shifted left by one or two positions depending on the round. For example- if the round is 1,2,9 and 16 the shift is done by only one position, For other round shift is done by two position.

Shifting

Rounds	Shift
1, 2, 9, 16	one bit
Others	two bits

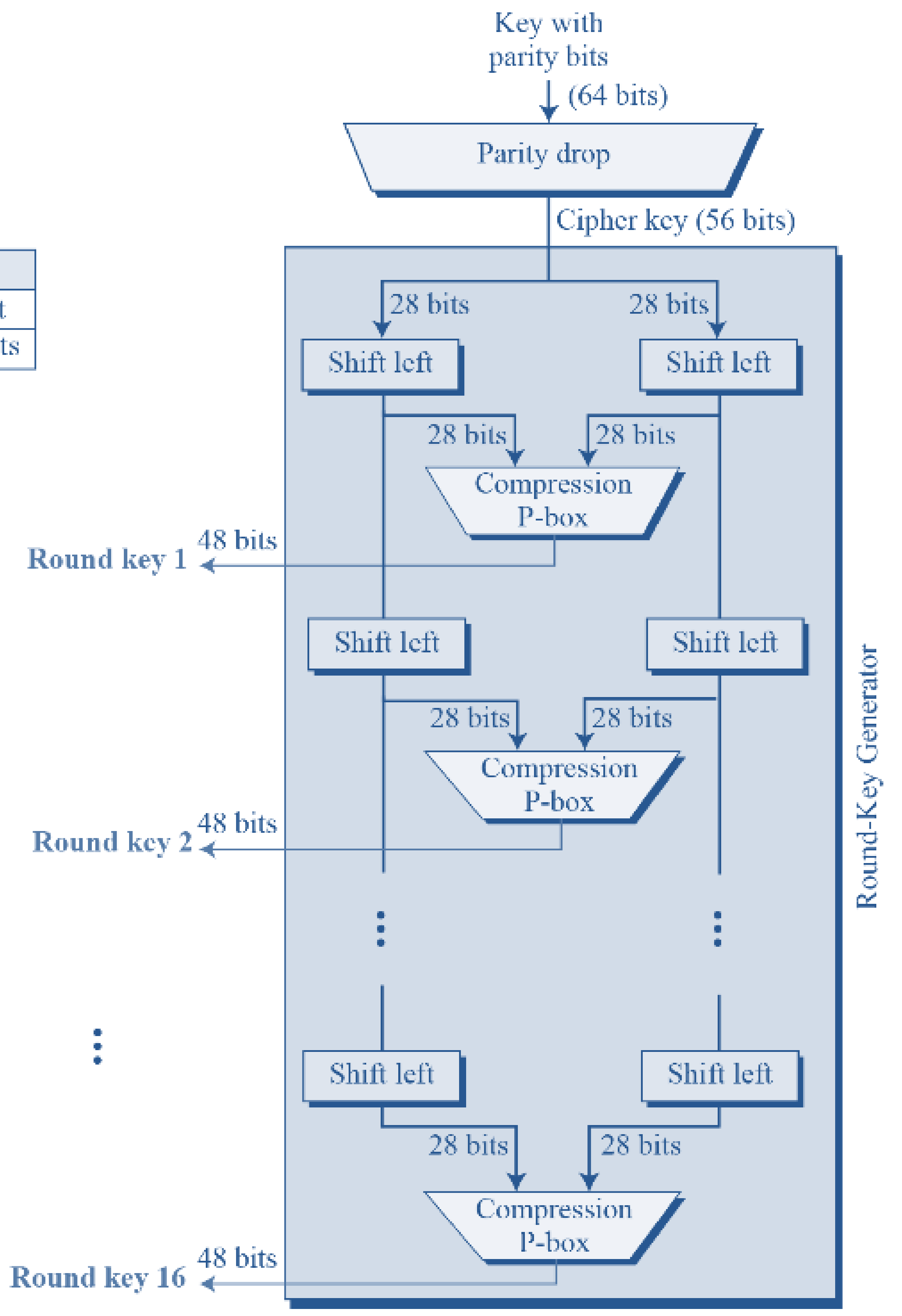


Fig. 3: Key Generation

Key Generation for DES

- After the appropriate shift 48 out of 56-bits are selected. For that composition permutation table is used. Because of this circular shift and compression permutation technique, a different sub key is generated for each round.

Shifting

Rounds	Shift
1, 2, 9, 16	one bit
Others	two bits

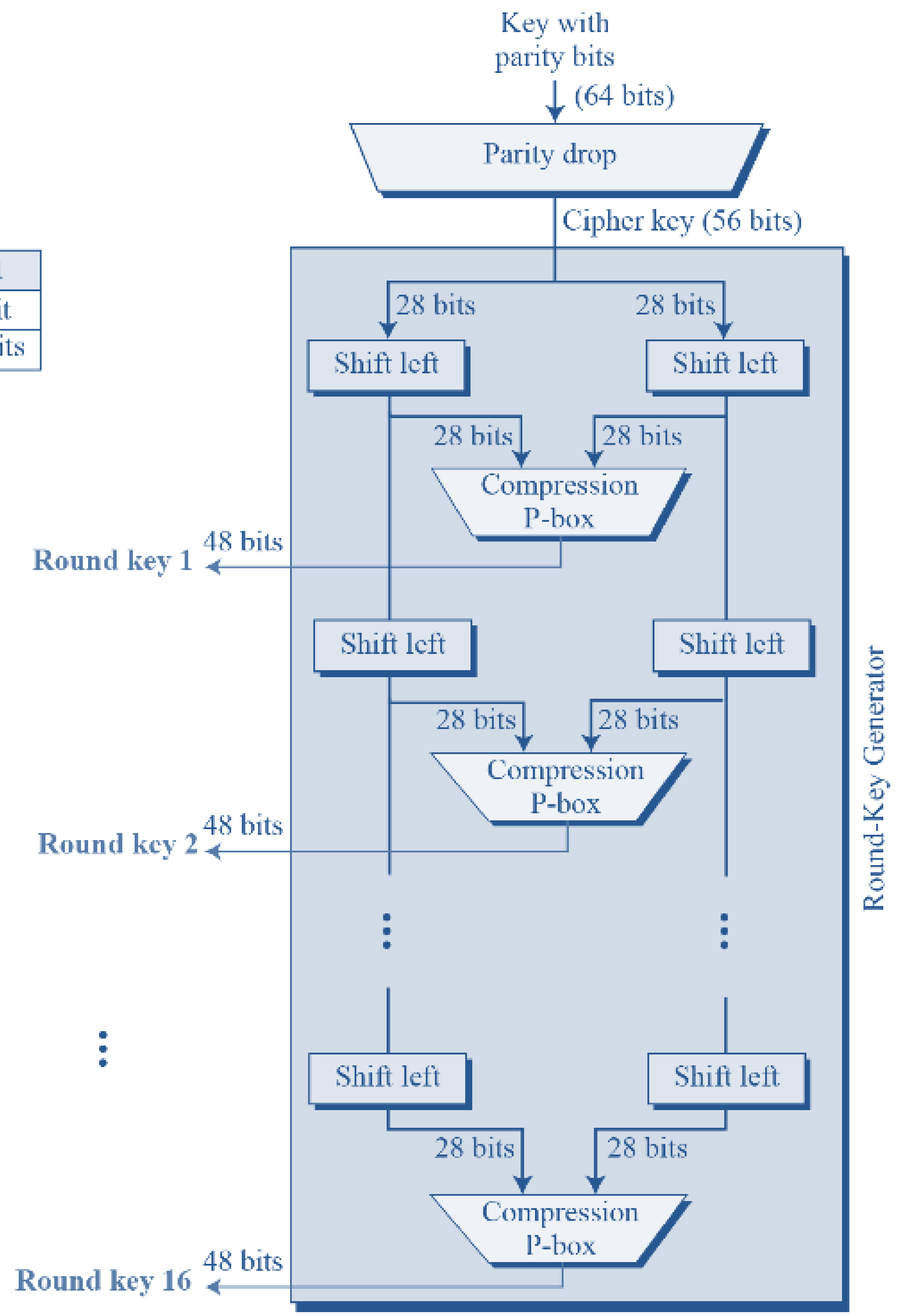


Fig. 3: Key Generation

Multiple Encryption and DES

- Multiple encryption is the process of encrypting an already encrypted message one or more times, either using the same or a different algirithm.
- It is also known as a cascade encryption or cascade ciphering or multiple encryption or superencipherment.

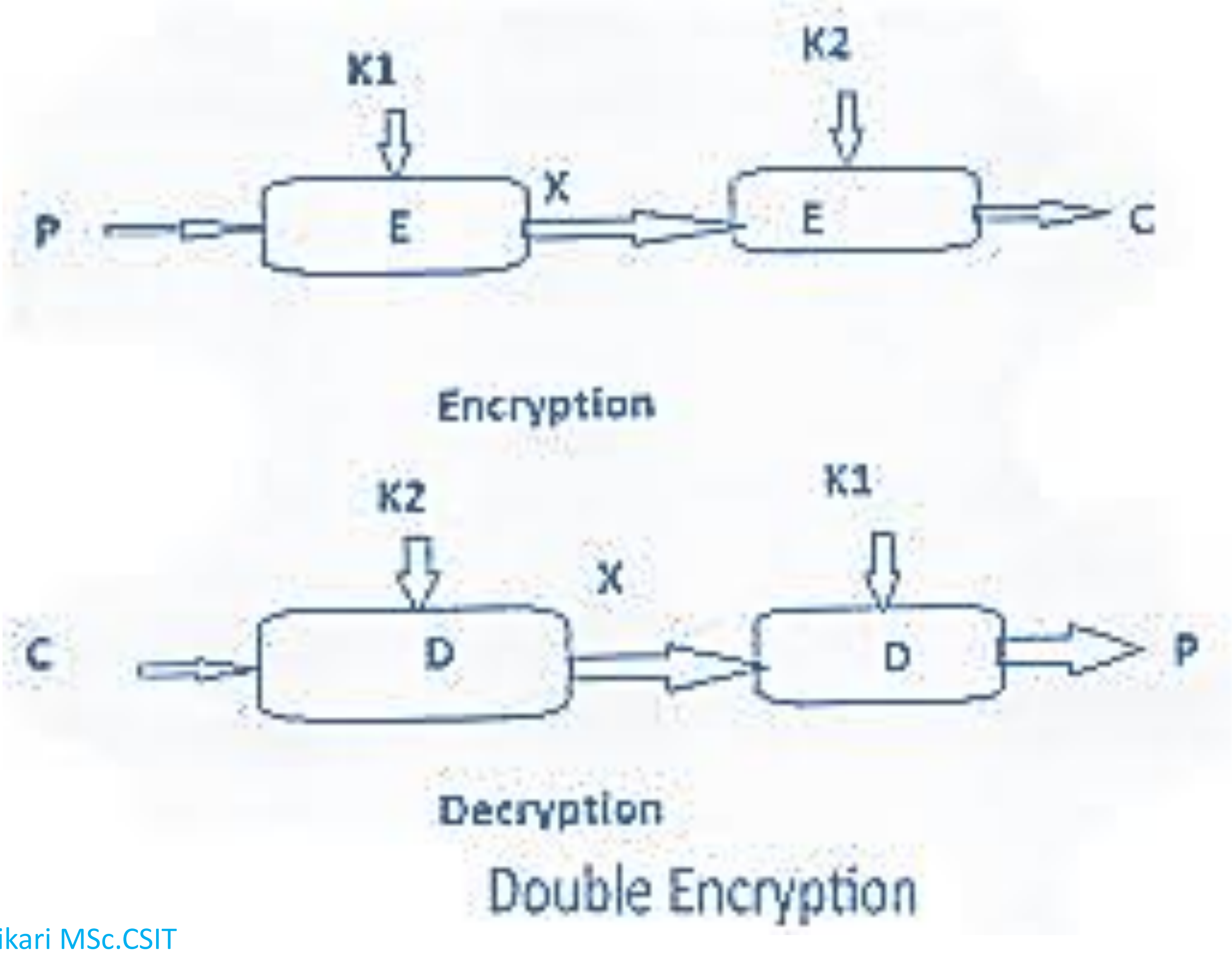
- Given the potential vulnerability of DES to a brute-force attack, there has been considerable interest in finding an alternative. One approach is to design a completely new algorithm, of which AES is prime example. Another alternative is to use multiple encryption with DES and multiple keys i.e. 2DES and 3DES.

1. Double-DES

- Double DES is a encryption technique which uses two instance of DES on same plaintext.
- In both instances it uses different keys to encrypt the plaintext. Both keys are required at the time of decryption.
- The 64-bit plaintext goes into first DES instance which then converted into 64-bit middle text using the first key and then it goes to second DES instance which gives 64-bit ciphertext by using second key.

i.e, $C = E_{k2}\ (E_{k1}\ (p))$

- However, 2DES can have the "meet-in-the -middle" attack which is also known as known as known plaintext attack.



2. Triple-DES

- Triple DES is a encryption technique which uses three instances of DES on same plaintext.
- It uses three different types of key choosing technique-
  1. All keys are different
  2. Two keys are same and one is different
  3. All keys are same
- Given a plaintext message, the first key is used to DES-encrypt the message.
- The second key is not the right key, this decryption just scrambles the data further.
- The twice scrambled message is then encrypted again with the first key to yield the final ciphertext.

i.e, $C = E_{k1}\ (\ D_{k2}\ (\ E_{k1}\ (\ P\ )\ )\ )$

$P = D_{k1}\ (\ E_{k2}\ (\ D_{k1}\ (\ C\ )\ )\ )$

- Similarly, triple DES can also be done with 3-separate key instead of only two.

i.e, $C = E_{k3}\ (\ D_{k2}\ (\ E_{k1}\ (\ P\ )\ )\ )$

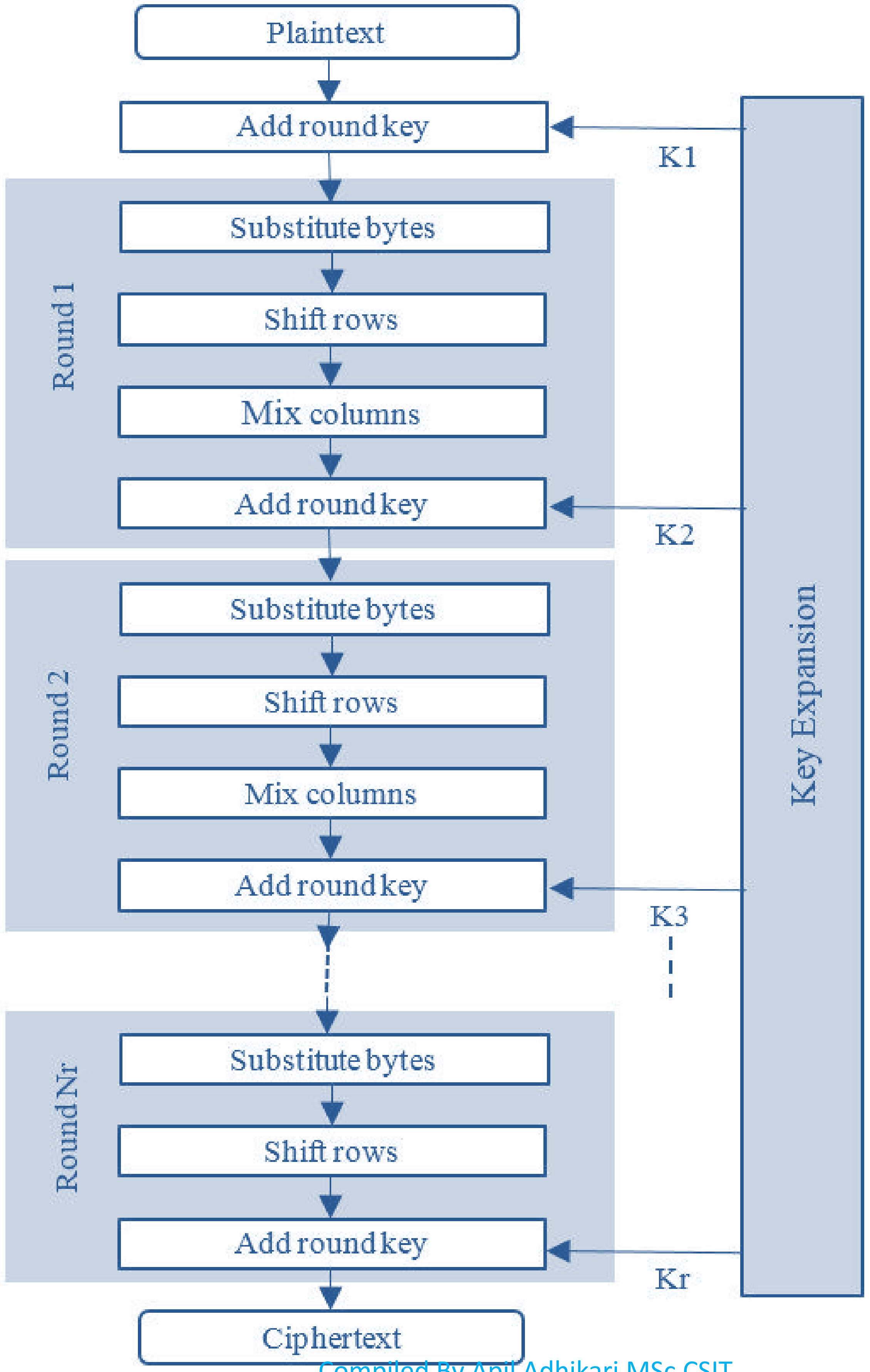
$P = D(K1, E(K2, D(K3, C)))$

Advance Encryption Standard (AES)

Features or key points of AES (Rijndael) :

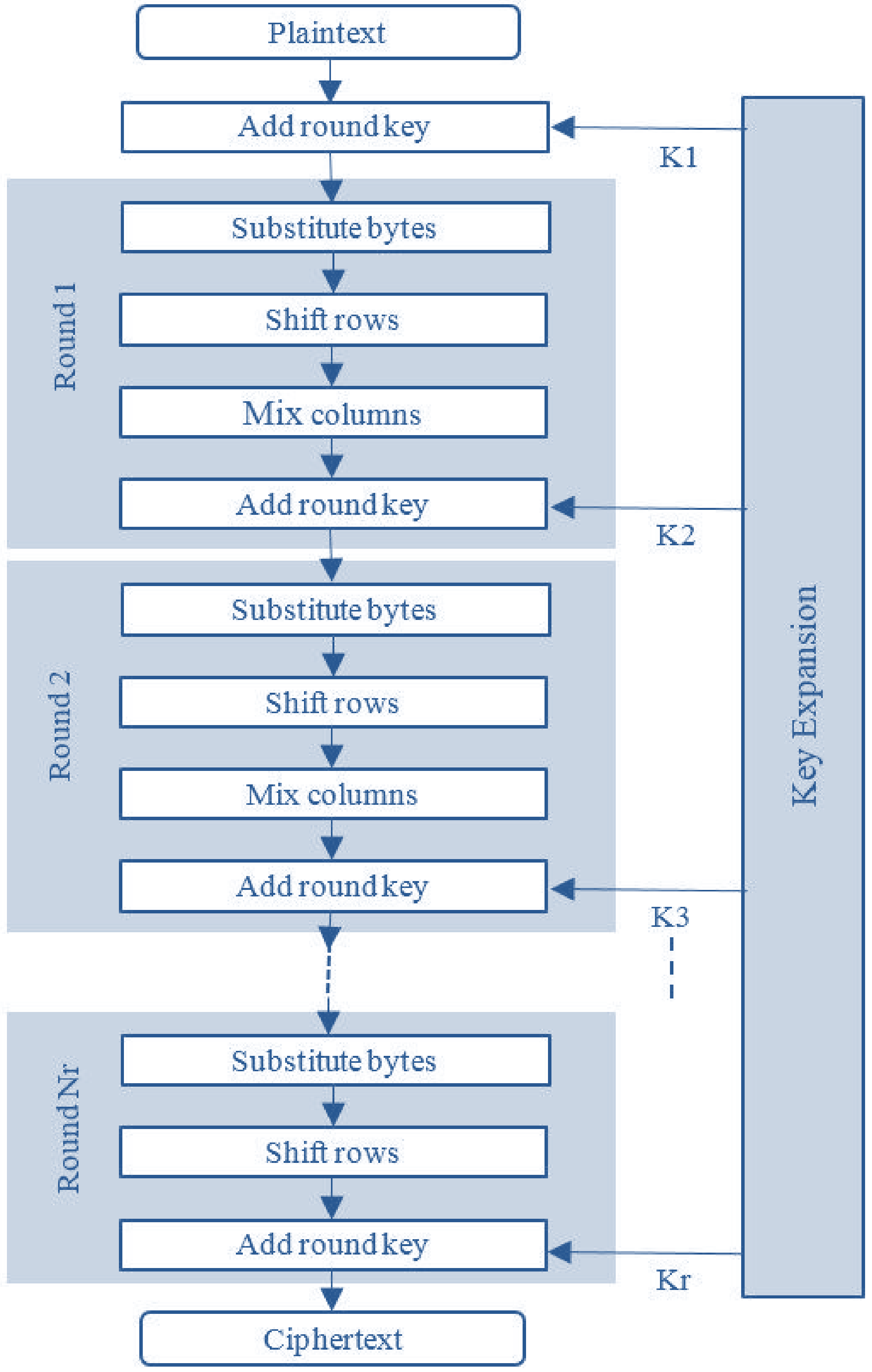
1. Key Length = 128, 192, 256-bits
2. Block Size = 128-bit
3. Cipher Type = Symmetric Block Cipher
4. Developed = 1998-2001 AD.
5. Possible key = 2^{128} , 2^{192} , 2^{256}
6. Round =
   i. 128-bit = 9+1 = 10 rounds
   ii. 192-bit = 11+1 = 12 rounds
   iii. 256-bit = 13+1 = 14 rounds

Advance Encryption Standard (AES)



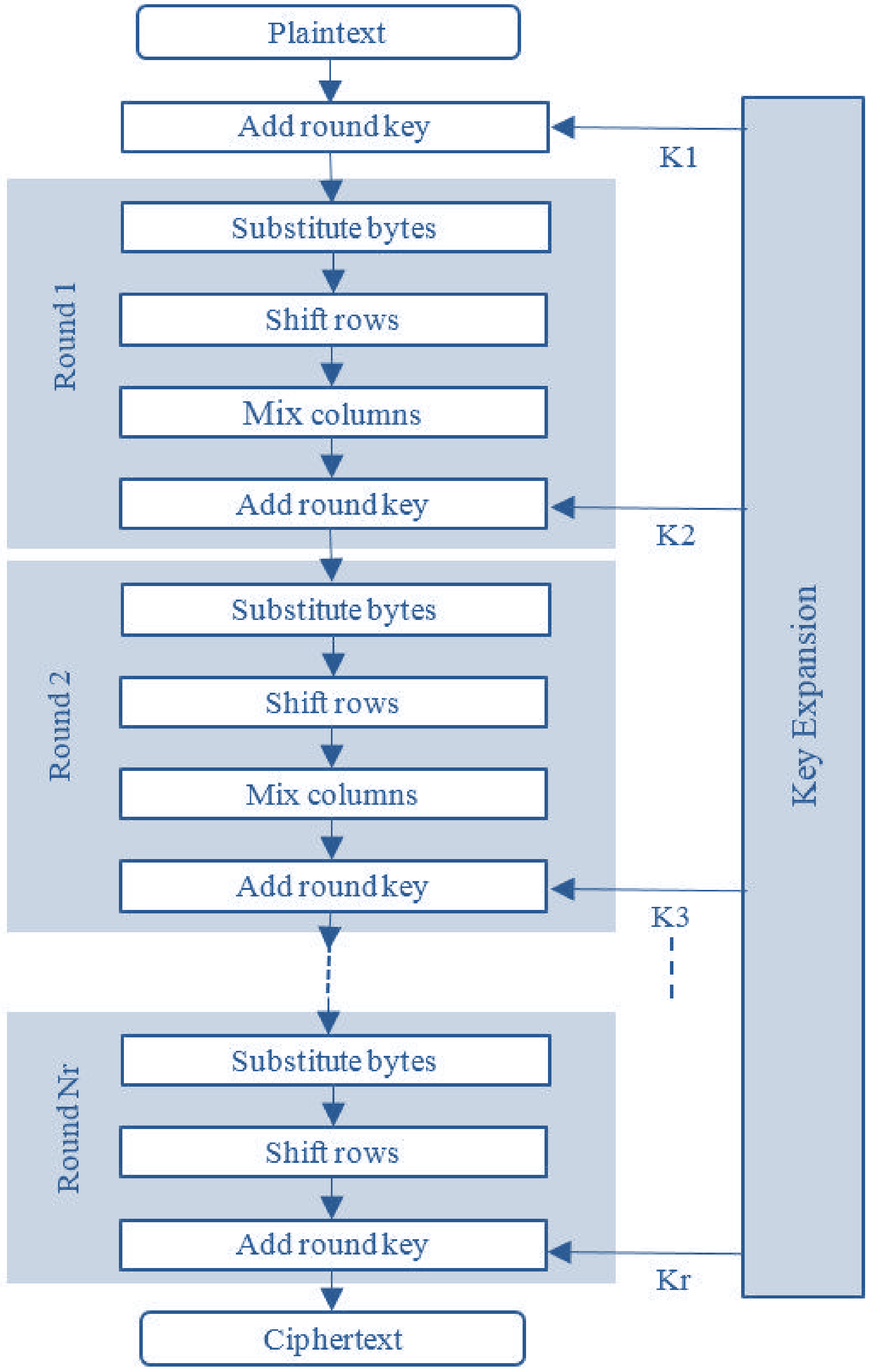
Advance Encryption Standard (AES)

- The overall structure of AES encryption process is shown in figure.
- The algorithm takes a plaintext block size of 128-bit or 16 byte.
- The key length can be 16, 24 or 32 byte (i.e. 128, 192 or 256 bit).
- The algorithm is referred to as AES-128, AES-192, AES-256 depending on the key length.
- The input to the encryption and decryption algorithm is a single 128 bit block. This block is considered as a 4x4 square matrix of bytes. The block is copied in to the state array which is modified at each stage of encryption or decryption.



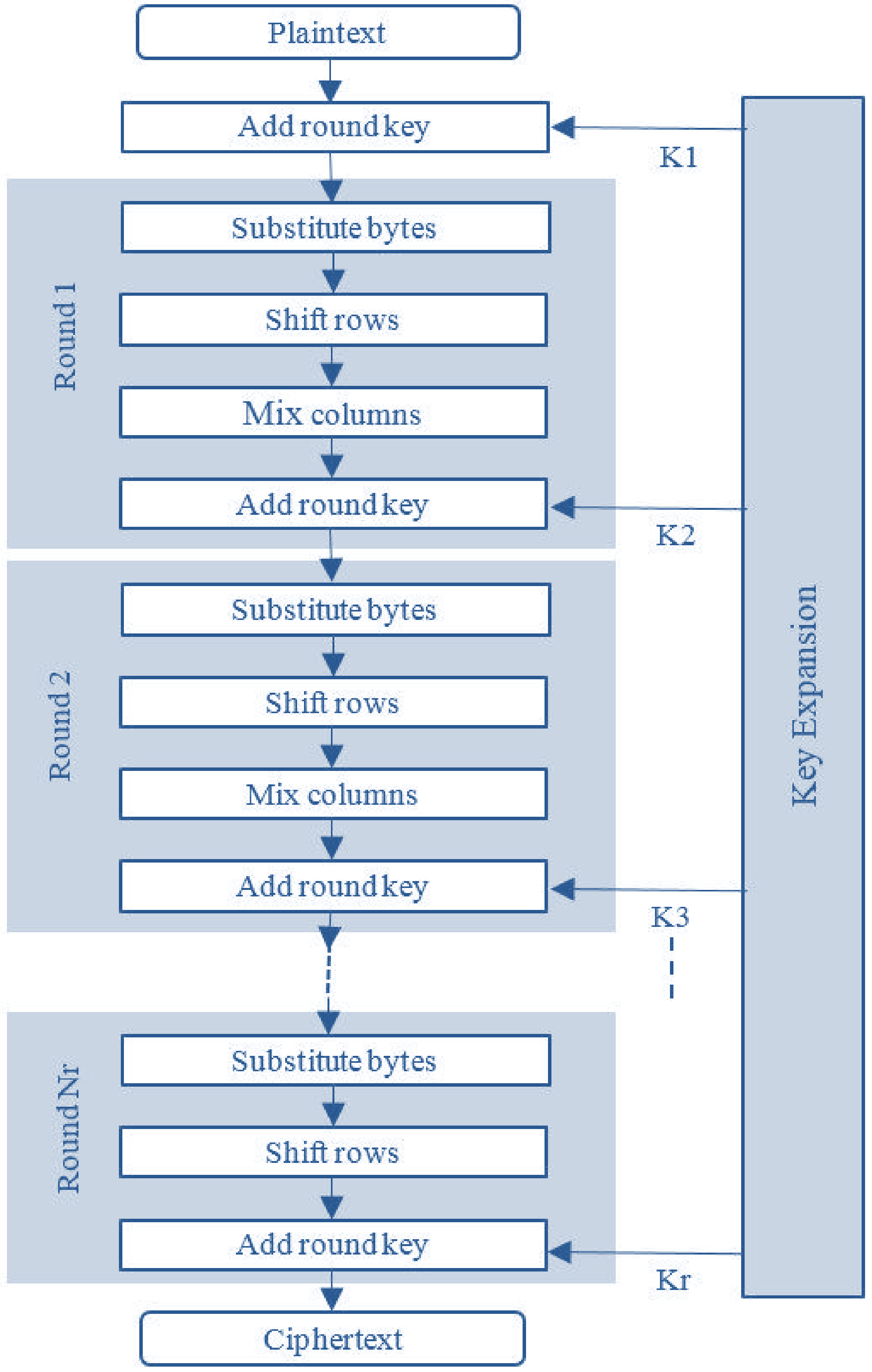
Advance Encryption Standard (AES)

- The cipher consists of N rounds, where the number of round depends on the key length; 10 rounds for a 16-byte key, 12 rounds for 24 byte key and 14 rounds for 32 byte key.
- The first N-1 round consists of 4-distinct transformation functions-
  1. Substitution byte
  2. Shift rows
  3. Mix columns
  4. Add round key
- The final round contains only three transformation and there is a single initial transformation (Add round key) before the first round which can be considered as a round 0.



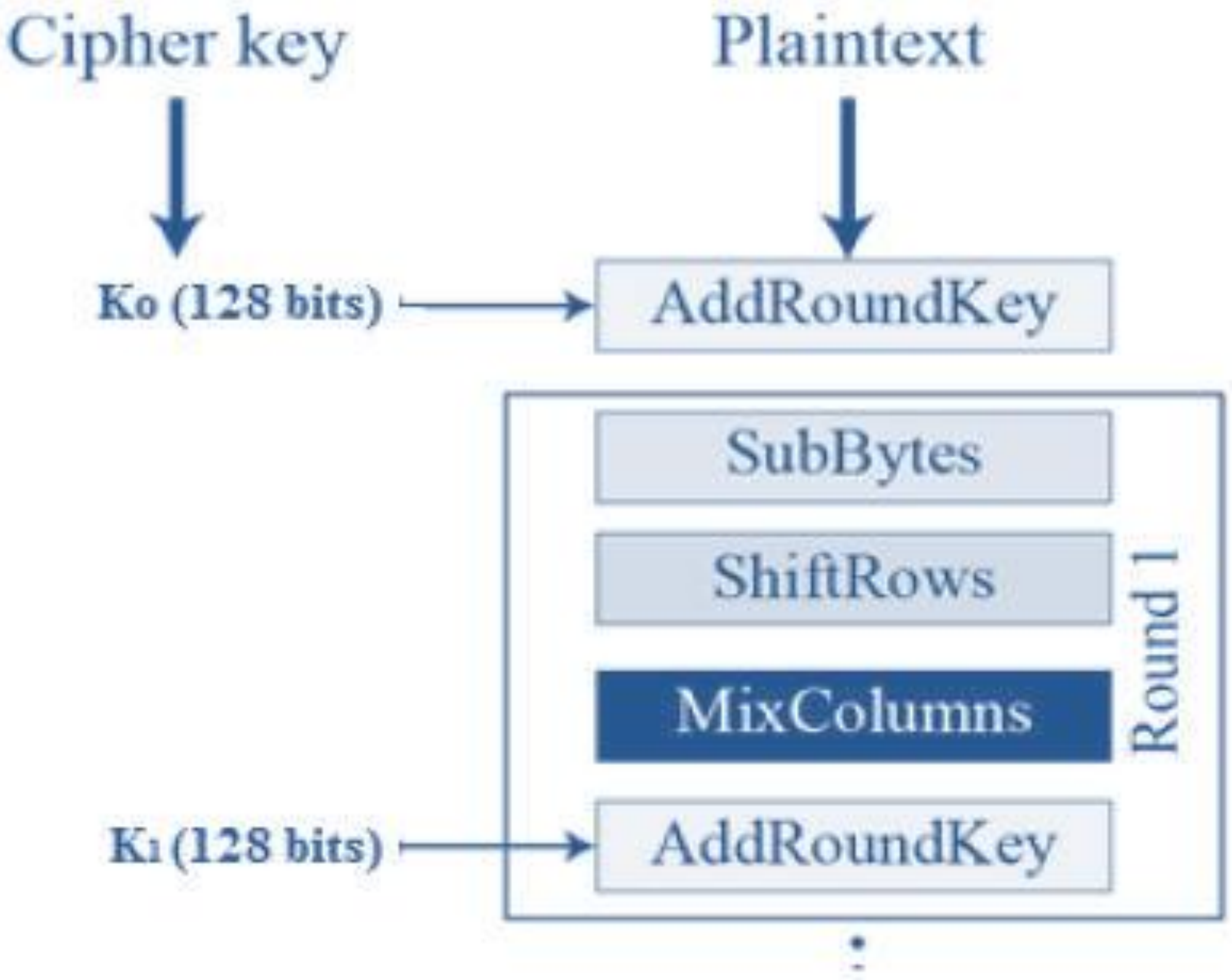
Advance Encryption Standard (AES)

- Each transformation takes 1 or more 4x4 matrices as input and produce a 4x4 matrix as output and final output of the final round is the ciphertext.
- Also the key expansion function generates N+1 round keys each of which is a distinct 4x4 matrix.



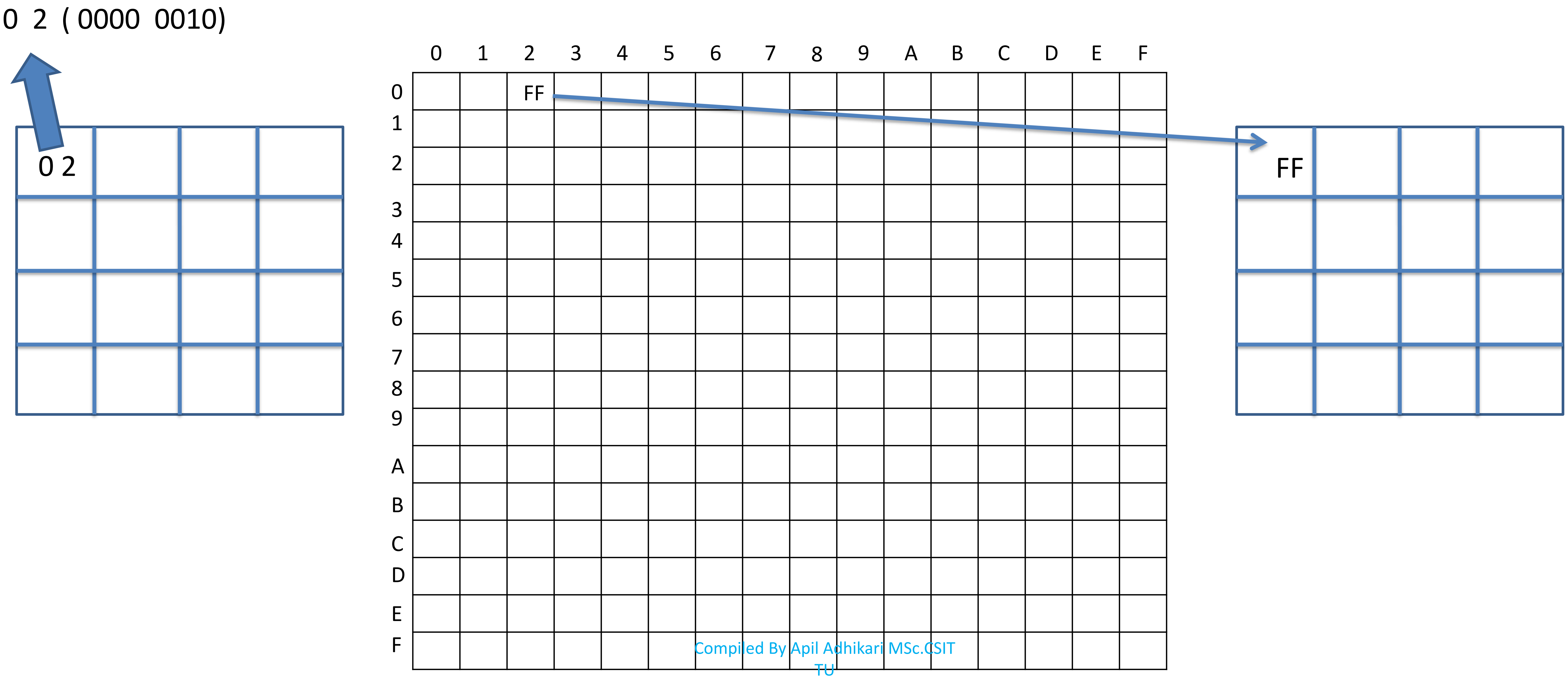
One round of AES

- In a single round of AES four different transformation operations are used out of which one in permutation and 3 are substitution.



1. Byte Substitution

- In this stage S-Box table is used to perform a byte by byte substitution of the block.



2. Shift Rows

- It is a simple permutation transformation.
- Each of the 4-rows of the matrix is shifted to the left.
- Number of Shift depends on the rows-

  1. First row is not shifted
  2. Second row is shifted one byte position to the left.
  3. Third row is shifted two byte position to the left.
  3. Fourth row is shifted three byte position to the left.

- The result is a new 4x4 matrix.

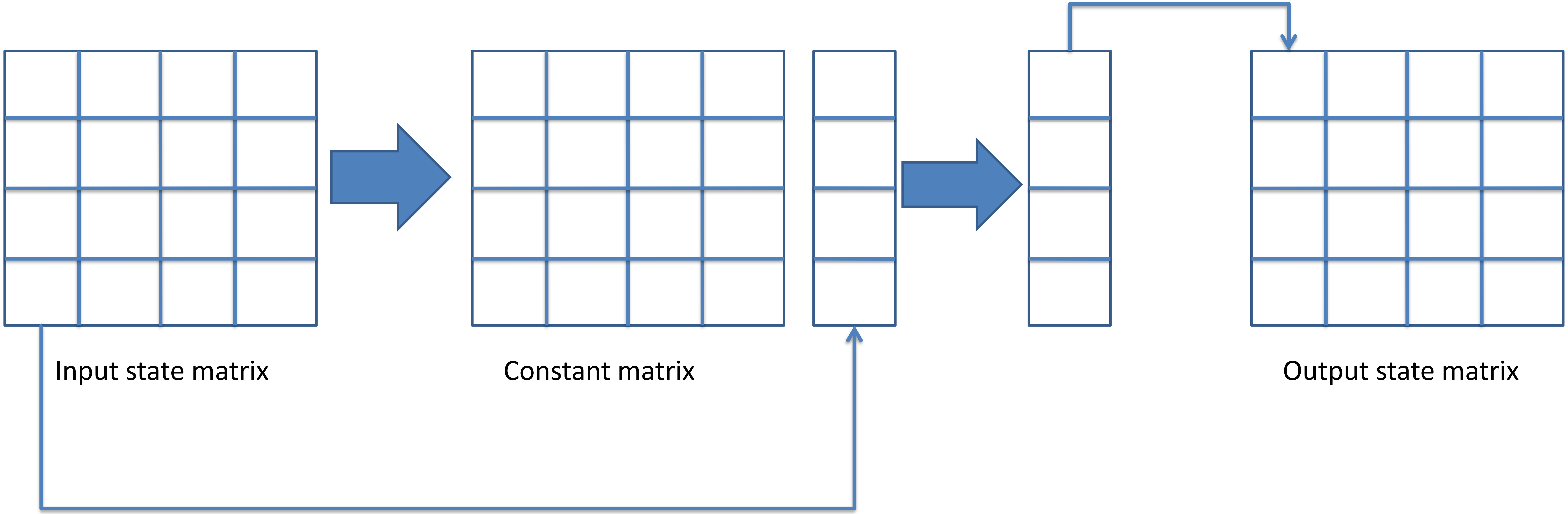
b1	b2	b3	b4
b5	b6	b7	b8
b9	b10	b11	b12
b13	b14	b15	b16

Shift Row

b1	b2	b3	b4
b6	b7	b8	b5
b11	b12	b9	b10
b16	b13	b14	b15

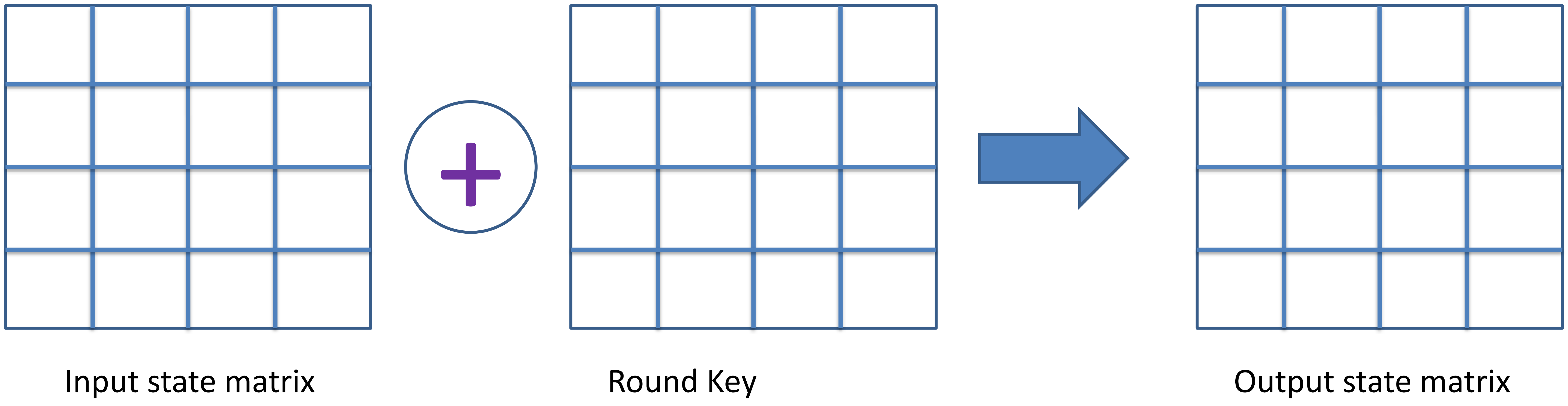
3. Mix Column

- Take each word/column i.e 4-bytes or 4x1 matrix and multiply it with the constant matrix.
- The output is a 4x1 matrix and is stored in a state matrix.



4. Add Round Key

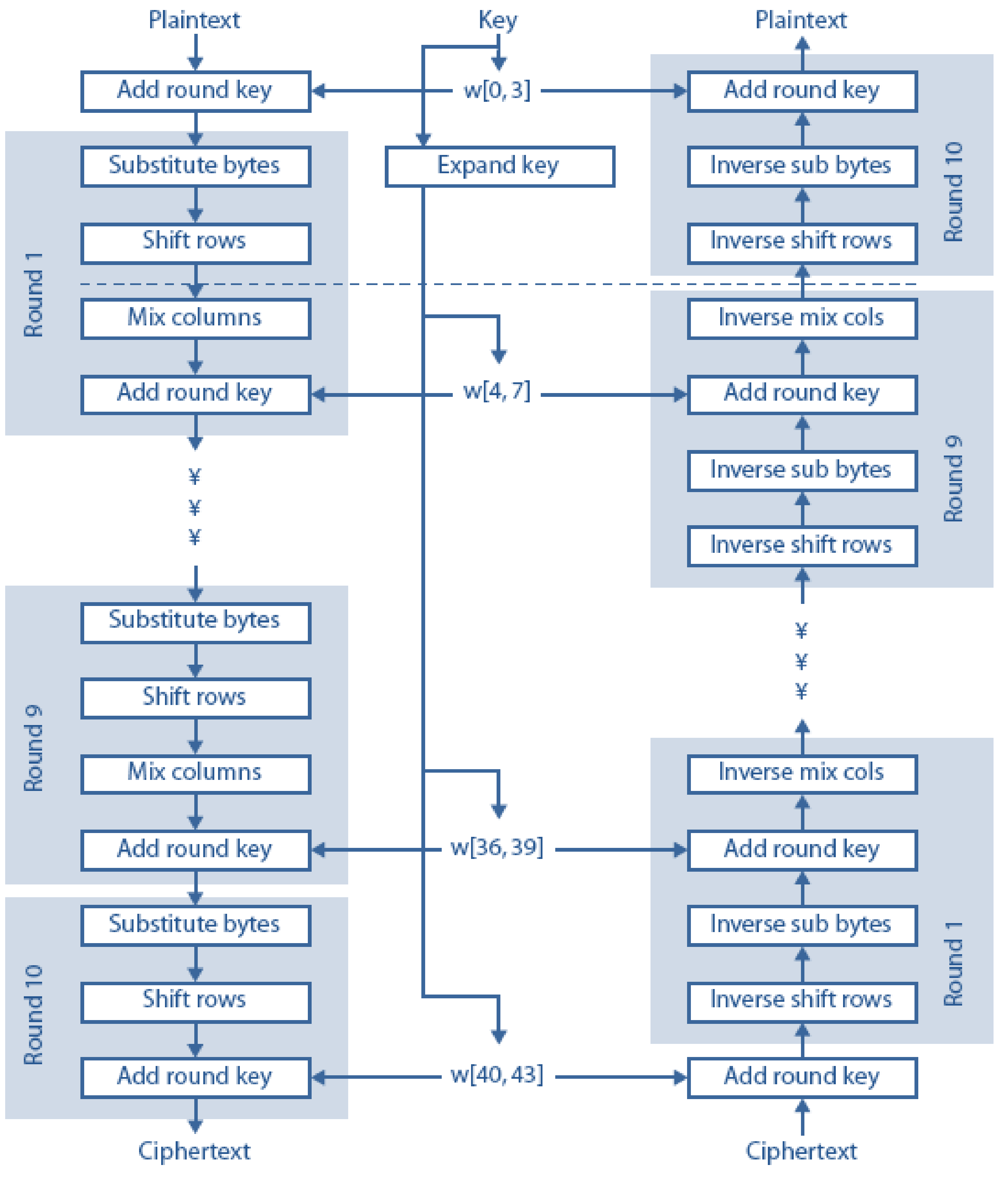
- The 16 bytes of the matrix are now X-ORed with the round key.



AES Decryption

- The process of decryption of an AES ciphertext is similar to the encryption process in the reverse order.
- Each round consists of the four processes conducted in the reverse order-

1. Add round key
2. Inverse mix columns
3. Shift rows
4. Inverse byte substitution



(a) Encryption
(b) Decryption

Key Generation for AES-128 bit

- For AES-128 bit encryption we have 128-bit key and which is represented by 4x4 matrix.

b1	b5	b9	b13
b2	b6	b10	b14
b3	b7	b11	b15
b4	b8	b12	b16
W_0	W_1	W_2	W_3

- For 128-bit AES we have 10 rounds and we need 11 different sub-key. So we need 11x4 = 44 words.

$$\underbrace{W_0, W_1, W_2, W_3}_{K_0}, \underbrace{W_4, W_5, W_6, W_7}_{K_1}, W_8, \ldots\ldots\ldots, \underbrace{\ldots, W_{43}}_{K_{10}}$$

Key Generation for AES-128 bit

- Initially we only have a 4- words. Whatever the words we have, we have to generate next 4- words from the previous 4-words. For that we have to follow following 3 steps -

1. Rotation Words (Rot word)

- For generating W_4 we need to use W_0 and W_3 .
- In this step we rotate W_3 in left direction (i.e. left circular shift).

2. Substitution Words (Sub word)

- It simply uses 16x16 matrix and substitute the value.

3. Round Constant

- For every round we have a different round constant values.

Key Generation for AES-128 bit

- Now next 4-words can be generated as-

$W_4 = W_0 \oplus$ Round Constant (Sub Word (Rot Word (W_3)))

$W_5 = W_1 \oplus$ Round Constant (Sub Word (Rot Word (W_4)))

$W_6 = W_2 \oplus$ Round Constant (Sub Word (Rot Word (W_5)))

$W_7 = W_3 \oplus$ Round Constant (Sub Word (Rot Word (W_6)))

!!!Thank you!!!